

INHA UNIVERSITY TASHKENT DEPARTMENT OF CSE & ICE

Fall 2023 – Application Programming in Java

Lab Assignment # 4

Topic covered : Exception handling

Last Date of Submission: 15/10/2023

INSTRUCTIONS:

- Program should be well commented.
- Program should be easy to use.
- All Lab assignments are to be completed by an individual student on his/her computer system and not in groups

Write a program that allows a user to input integer values into a 10-element array and search the array.

- The program should allow the user to retrieve values from the array by index or by specifying a value to locate.
- The program should handle any exceptions that might arise when inputting values or accessing array elements.
- The program should throw a **NumberNotFoundException** if a particular value cannot be found in the array during a search. If an attempt is made to access an element outside the array bounds, catch the `ArrayIndexOutOfBoundsException` and display an appropriate error message.
- Also, the program should throw an `ArrayIndexOutOfBoundsException` if an attempt is made to access an element for which the user has not yet input a value.

Note: You have to implement class **NumberNotFoundException**.